

Designing a Cyber Situational Awareness Application for Space Missions

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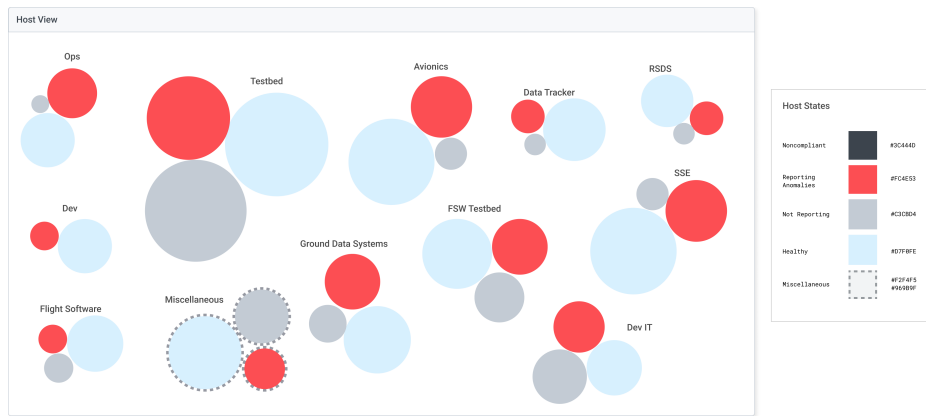


Figure 1: The host dashboard of CyberSAM: a situational awareness application for space missions. Hosts are visualized as bubbles and clustered into mission-specific categories relevant to operators. Color denotes the state of each host, while the bubble diameters correlates with the volume of events. This visualization facilitates a rapid assessment of the overall mission ground system posture.

Abstract

Effective cyber situational awareness is crucial for protecting critical space missions from advanced threats. We developed a novel capability called CyberSAM at the Jet Propulsion Laboratory that integrates cyber threat detection with mission objectives, stakeholder needs, operational constraints, and spacecraft ground system use cases. This enables mission operators to quickly receive relevant insights for decisive action, enhancing mission success and increasing response time during critical events. CyberSAM's core

innovation lies in its systematic Human-Centered Design (HCD) process, which engineers a mission-centric cyber situational awareness system that trades complexity for usability. By prioritizing user experience, CyberSAM provides an intuitive interface that reduces cognitive load for operators, achieving a 31% reduction in cognitive workload, reduces context-switching costs by 40%, and measurably improves incident response time during mission operations. The implementation of a scalable design system further reduced development time by 38% when onboarding new missions, ensuring consistency across mission applications while accelerating deployment. This case study discusses our insights building CyberSAM with a human-centered approach, highlighting the challenges encountered, measurable operational gains, and lessons learned in developing a cyber-defense solution for high-stakes space missions.

CCS Concepts

• **Human-centered computing** → **Field studies**; • **Security and privacy** → *Domain-specific security and privacy architectures; Usability in security and privacy.*

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1 Introduction

A well-defined situational awareness strategy is key to enabling effective defense against the cyber adversary. For space missions, the challenge lies in engineering a defense strategy that is transparent and unobtrusive to the main objectives of the mission itself. The act of simply deploying commercial-off-the-shelf (COTS) cybersecurity tools into the mission environment has risks and limitations. Blackbox COTS products lack mission context and are often not adapted to mission-specific requirements. An effective mission-centric strategy requires an understanding of mission objectives, stakeholder needs, operational constraints, and spacecraft ground system use cases. We present a case study of *CyberSAM*, a novel situational awareness capability developed by JPL cyber research for use by space missions. We frame this work as a deployed case study to highlight the practical application of Human-Centered Design (HCD) in a high-stakes, mission critical environment. Unlike theoretical models, *CyberSAM* has been piloted as cyber defense solution for protecting the ground systems for several NASA missions, including Mars missions, missions to Jupiter's moon Europa, Earth missions, and the Deep Space Network. It has been instrumental in maintaining mission asset inventories, detecting and diagnosing anomalous user and host behavior, identifying firewall misconfigurations, and ensuring compliance with NASA-required NIST 800-53 controls.

CyberSAM's key innovation is applying a systematic HCD process to engineer and optimize a mission-centric cyber situational awareness system. This approach allows us to trade complexity for usability, providing a cyber-defense solution woven into the mission environment, respecting mission constraints, tuned to mission context, and designed for the needs of mission operators. *CyberSAM* includes a variety of algorithms (rule-based, heuristic, algorithmic, and ML-based approaches) for anomaly and threat detection, implemented in the Splunk query language [18], Splunk machine learning toolkit, and Python. The UI elements are based on React, and the entire toolset was deployed as an app on the JPL Splunk infrastructure. In this paper, we present a detailed overview of the challenges in designing a situational awareness application for space missions and document the important lessons learned.

2 Background and Related Work

Designing for space operations requires navigating a highly complex and multidisciplinary domain where standards for end-to-end security usability are scarce. This section reviews the intersection of Secure Systems Engineering, Human-Computer Interaction

(HCI) in space, and Cyber Situational Awareness. The rapid expansion of the space threat landscape, now characterized by sophisticated cyber-enabled attacks against both ground and space segments [8, 13], necessitates a shift from traditional security models to human-centered defense strategies [19]. As the domain of *SpaceCHI* evolves, research highlights that security tools must be adapted to the unique cognitive constraints of space exploration, where standard enterprise solutions often fail to support operator workflows during critical mission phases [14]. For example, the *AESOP* (Adaptable Environment for Space Operations) [5] study demonstrated that applying User-Centered Design (UCD) to satellite command and control interfaces significantly reduced operator workload and improved anomaly resolution times compared to legacy systems. Our work extends UCD principles specifically into the domain of cyber defense, where cognitive vulnerabilities [3] are as critical as software vulnerabilities.

Situational Awareness (SA) involves the perception of elements in the environment, the comprehension of their meaning, and the projection of their status in the near future [7, 12]. Modern cyber situational awareness frameworks are moving beyond simple monitoring to include predictive state projection [16] and mission-oriented data evaluation [17]. However, existing visualization literature suggests a persistent gap: most tools remain tailored for security experts rather than domain operators [15]. In the context of cybersecurity, Cyber Situational Awareness (CSA) requires operators to not only see data but to understand the implications of that data regarding threats and mission health. This is complicated as often there is no clear understanding of the different stakeholders for CSA visualizations. [10] Research has long shown that “security is not a primary task” for most users [21], and in space missions, the primary task is science and telemetry, not security logs. Bridging this gap requires the integration of explainability to ensure transparency and trust in automated decision-making [2, 9], aligning with emerging strategic criteria for robust, actionable situational awareness [4]. *CyberSAM* addresses this gap by combining detection algorithms with a human-centered visualization layer, prioritizing the operator's ability to contextualize cyber security information within a mission context.

3 Operational Challenges

In this section, we discuss the challenges faced while designing and deploying our situational awareness application for space missions. This includes situational awareness challenges as well as architectural constraints. A primary challenge was defining what “Situational Awareness” actually means for a space mission operator versus a traditional Security Operations Center (SOC) analyst. In a mission context, SA is not merely about aggregating logs; it is about *contextualization*. In a mission environment, data is meaningless without the operational context that frames it; a spike in network traffic might be a data exfiltration attempt, or it could be a scheduled downlink of science telemetry. The system must enable the operator to distinguish between these scenarios instantly. This requirement for contextualization introduced two specific challenges.

Integration into the mission workflow: For a cyber-defense solution to be successfully adopted, it must be woven into the fabric of the mission environment. It must respect mission constraints,

be tuned to the mission context, and be designed specifically for the needs of mission operators. Traditional cybersecurity tools are frequently dismissed in these environments because they employ obscure security vocabulary, lack an understanding of mission phases (e.g., launch vs. cruise), and generate “noise” that adds to the cognitive load of operators already managing complex spacecraft systems. To avoid these pitfalls, CyberSAM was designed to fit directly into the mission workflow, delivering relevant, actionable insights in a way that resonates with mission users and addresses their specific operational cadence.

Explainability and Transparency: In mission-critical ground data systems, rapid and accurate decision-making is paramount. Operators require a transparent understanding of the underlying logic behind their tools; “black-box” solutions are inadequate for safety-critical environments where trust is essential. Therefore, a cyber defense solution must prioritize explainability, enabling operators to manually verify and assess automated results. Beyond issue alerts, the system must provide granular context—such as the anomaly’s nature, triggering data points, and observed patterns. This level of detail empowers operators to swiftly validate threats and execute informed response actions.

Beyond the SA challenges, there were several architectural challenges that constrained our design.

Evolving and incomplete requirements: CyberSAM was developed to support a wide range of tasks in a domain where cybersecurity monitoring requirements were initially undefined. To address this, the team anchored foundational requirements on the widely used MITRE ATT&CK framework [20]. However, as the tool evolved from a research prototype to a deployed solution, new requirements were continually introduced. This constant evolution made maintaining a cohesive architecture a significant challenge.

Multiple customers (missions) each with unique requirements: Space missions often involve multiple stakeholders, each with their own unique security requirements and operational priorities. Cyber situational awareness must be designed to support a diverse array of customers and missions, ensuring that each receives the necessary level of protection without compromising the others. Balancing these competing demands requires a nuanced understanding of each mission’s specific needs while maintaining a unified situational awareness framework that can cater to all.

Diverse use cases: CyberSAM originated as a targeted anomaly detection tool for space mission ground systems. However, its scope rapidly expanded to encompass sophisticated threat detection, operational health diagnostics, and compliance verification supporting NIST 800-53 controls for NASA’s Assessment and Authorization (A&A) process. This functional broadening presented significant challenges in maintaining architectural cohesion. Consequently, the platform underwent extensive iterative development between versions 1 and 4, necessitating comprehensive redesigns of both the frontend and backend to accommodate these heterogeneous requirements within a unified framework.

Extensibility and Adaptability: Given the dynamic nature of the cyber threat landscape, the system had to be modular. On the backend, this meant supporting seamless integration of new data sources and detection algorithms without overhauling the core infrastructure. On the frontend, adaptability was vital to accommodate diverse operator needs, allowing for personalized dashboards

and mission-specific visualizations. This modularity ensures the system can evolve with technological advancements and changing mission objectives.

Support both research and production: A comprehensive cyber defense solution must effectively bridge the gap between research flexibility and production stability. The research component requires sandboxed environments and data pipelines to foster rapid prototyping of new detection algorithms. In contrast, the production environment prioritizes reliability, scalability, and real-time processing to meet mission-critical performance standards. Supporting both modes creates a continuous feedback loop where research innovations are rigorously validated before operational deployment. This approach ensures the system continually evolves to address new threats without compromising the integrity of live mission systems.

4 Design Methodology

CyberSAM was developed using a Human-Centered Design (HCD) approach tailored to mission-centric environments. The process involved user research, data synthesis, and iterative design and development, with decisions continuously guided by insights from stakeholders to ensure the application met both user needs and the unique demands of cybersecurity operations. To understand the requirements of multiple stakeholders, user research included interviews with five mission operators responsible for operational assurance of spacecraft and ground systems and seven cybersecurity team members, all serving as subject matter experts (SMEs), complemented by a heuristic evaluation. These SMEs were selected across five different space missions to ensure diverse operational representation. The heuristic evaluation, based on Jakob Nielsen’s ten usability principles, identifies potential usability issues by analyzing user interactions against established design guidelines. Data from semi-structured interviews and usability evaluations were analyzed to identify user and mission needs, validate design assumptions, and inform data-driven decisions. This analysis guided the development of an intuitive application that reduces cognitive complexity and supports effective performance in high-stakes mission environments.

5 CyberSAM System Design

To reduce complexity in the CyberSAM application, a human-centered design approach emphasized three user-centered design principles: usability, consistency, and scalability. Usability and consistency enhance user efficiency, while scalability improves efficiency for designers and developers.

5.1 Design for Usability

The first user-centered design principle, designing for usability, emphasizes ease of understanding and navigation, including accessibility, efficiency, and learnability. User interviews and heuristic evaluation revealed that the existing navigation menu did not meet these standards, as it was cluttered, repetitive, and unable to scale to support CyberSAM’s expanding range of missions and capabilities, limiting its effectiveness across diverse use cases. To address these limitations, redundancy was eliminated and the information

architecture was reworked for scalability and flexibility. This redesign enabled CyberSAM's evolution from anomaly detection in space mission ground systems to advanced threat detection, operational health monitoring, diagnostics, and NIST 800-53 compliance within NASA's A&A process, while maintaining usability across both frontend and backend systems. By accommodating multiple workflows and mission contexts, the redesigned structure supports diverse use cases. Further, by grouping hosts into mission-relevant categories (see Figure 1), the CyberSAM dashboard provided the operational context necessary for operators to quickly assess the security posture of the ground data system. This visualization enhanced explainability by integrating cybersecurity monitoring directly into daily mission-centric workflows.

5.2 Design for Consistency

The second user-centered design principle, designing for consistency, enhances usability and learnability by enabling users to navigate the application efficiently and reducing confusion. This principle is paramount in mission-critical environments such as space missions. Because mission operators are focused on the primary objective of ensuring mission success, they require clear, accurate, and actionable insights from their tools. A heuristic evaluation of CyberSAM revealed inconsistencies in terminology, page layouts, data visualizations, and color usage. To improve coherence and reduce complexity, standardization was applied across the application.

Standardization for Terminology: Terminology was standardized to address inconsistencies, particularly in anomaly severity, which had been represented both numerically (0–4) and as text (informational, low, medium, high, critical) across the UI and code. Standardizing severity as text ensures that encoded information can be naturally read by people, providing clear distinction and reducing the risk of confusion or misinterpretation. This clarity enables users to quickly understand severity levels and make rapid, accurate decisions when responding to anomalies.

Standardization for Data Visualizations: Because each mission collects its own specific data, visualizations are a crucial part of the CyberSAM application, enabling operators to identify patterns relevant to their mission context accurately and without ambiguity. To support clarity and usability, charts were designed to be simple, consistent, and modular. A library of modular charts, informed by user interviews and heuristic evaluation, allows users to quickly interpret patterns and investigate anomalies. Most charts were drawn from the Splunk library to streamline development, with a single custom chart created to meet a unique need for displaying anomaly distributions across hosts (see Figure 1).

Standardization for Color: Color plays a critical role in data interpretation. In CyberSAM, colors represent system states—red for urgency, yellow for warning, and blue for health. To ensure clarity, usability, and compliance with Section 508—which requires federal agencies to make Information and Communication Technology (ICT) accessible to people with disabilities, including those with color blindness—the updated color palette was streamlined and standardized across all visualizations, including the custom host chart created to display anomaly distributions (see Figure 1). Existing

colors were retained where users had already learned their meaning, and thorough testing confirmed sufficient color differentiation, readability, and accessibility for all users.

Overall, standardization across terminology, data visualizations, and color was critical to improving the usability and learnability of CyberSAM. These design decisions reduced ambiguity and cognitive load, enabling mission operators to quickly and accurately interpret system states, detect anomalies quickly, and increase response time during critical events. Based on findings from heuristic evaluation and user interviews, this consistency ensures the application effectively supports time-sensitive mission operations.

5.3 Design for Scalability

The final user-centered design principle, design for scalability, reduces implementation complexity and supports ongoing development. Scalability enhances flexibility, performance, and reliability, allowing CyberSAM components to evolve and new features to be added without reorganizing existing structures. This principle is critical for cyber situational awareness, as the application must constantly accommodate new detections and visualizations to address an evolving threat landscape. By creating a modular design system with reusable components and complementary front-end code, the application supports extensibility, enabling new data sources and detection algorithms to be integrated seamlessly, and adaptability, allowing personalized dashboards and mission-specific visualizations. This consistency reduces users' learning time, increases trust and engagement, and improves developer efficiency by minimizing rework and streamlining design and development.

Across these design principles, continuous usability testing with mission operators provided evidence that cognitive load was reduced by 31% over successive iterations. User feedback and heuristic evaluation indicated improved navigation efficiency, clearer interpretation of system states, and fewer errors during task execution, demonstrating that the application's increasing functionality did not come at the expense of situational awareness. These outcomes directly informed the lessons learned from deploying CyberSAM in mission-critical environments.

6 Lessons Learned

This section reflects on the development and deployment of CyberSAM across several NASA missions to distill key lessons learned regarding the application of HCD in high-stakes, mission-critical environments.

Early integration of HCD reduces downstream friction. Integrating HCD practices [12] early in the product life cycle fundamentally altered the system architecture. By understanding the operator's workflow *before* codifying the backend logic, we avoided building tools that were operationally unusable. This early alignment significantly improved user experience and increased product adoption among non-security experts.

Continuous usability testing as a form of risk management. To mitigate cognitive friction, continuous usability testing of the application [1] supported an iterative process aligned with evolving mission requirements. Task-based sessions, heuristic evaluations, and scenario-driven simulations assessed cognitive workload and

performance across iterations, ensuring that new features did not compromise situational awareness.

Leverage familiarity to lower cognitive load. Space mission operators are highly trained specialists with established mental models. By incorporating these pre-existing learned behaviors into CyberSAM, we created intuitive interfaces that required minimal retraining. Furthermore, using existing Splunk UI libraries ensured consistency with other tools the operators might already use, reducing the “context switching” cost by **40%**.

Standardization drives scalability. The creation of a modular design system was critical for supporting multiple missions. By standardizing terminology, layouts, and visualizations [6], we reduced the development time by 38% for onboarding new missions. This lesson reinforces that consistency is not just an aesthetic choice but an operational necessity for scaling security tools across diverse environments.

HCD reduces research-to-production lag. Evaluating designs with users proved to be a cost-saving measure, often saving hours of expensive backend development. [11] By validating our prototypes, we were able to make necessary improvements before committing to expensive frontend and backend development. This seated HCD as a bridge that facilitates a smooth transition of research capabilities into production environments while saving costs.

7 Conclusion

This case study demonstrates the operational value of applying human-centered design for cyber situational awareness in mission-critical space missions. While our findings underscore the strengths of this approach, they also emphasize the critical need for deep, ongoing engagement with specific mission contexts. The resulting design improvements not only enhanced usability and situational clarity but also measurably improved incident response time during mission operations. Future applications should build upon these lessons to refine methodologies that enhance the operator’s experience. By incorporating these insights into future security tools, we can contribute to a more robust understanding of user expectations and behaviors in high-stakes environments. Ultimately, this process of continuous reflection and adaptation serves to advance both theoretical knowledge and practical application within the field.

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